

K874 — Casey’s steer “linear algebra, one D_{IV}^5 domain” DISSOLVES the morning’s coordinate mess. The $k \leftrightarrow \nu$ dictionary, the Bergman-weight ladder, the two-point across-domain distance (2.666 nats, FK book), the nested-domain tower — all of it was IMPORTED multi-domain machinery, and it’s the source of the K872 coordinate inconsistency. One domain, one Hilbert space $H^2(D_{IV}^5)$, one 3×3 mass matrix $M_{ij} = \langle \psi_i | \hat{O} | \psi_j \rangle$ (= F585). Masses = eigenvalues. Diagonalize it. The three generations are three VECTORS (at the three ρ -strata of the ONE domain), not three domains. The real question, now sharp: what is the condensate operator $\hat{O}$ on H^2 , and what is its rank + spectrum? (Rank-1 $\rightarrow$ one mass, K869 — still the crux.) This SUPERSEDES my K872/K873 (the coordinate dictionary was patching an artifact).

Keeper | 2026-07-24 Fri | Casey’s standing linearization order, applied — and it retires my own multi-domain notes. Own it. Plain.

Casey’s steer (verbatim)

“linear algebra, one D_{IV}^5 domain.” (Standing order — feedback_linearization_standing_order; steered 4+ times this program.)

What it DROPS (imported multi-domain machinery = the coordinate mess)

- The nested-domain tower $D_{IV}^5 \supset D_{IV}^3 \supset \text{point}$ (my K852/K857 framing) — 3 domains, 3 genera, 3 Bergman weights.
- The Bergman-weight ladder across weighted spaces H_k (the $k=1,2,3$ / α -ladder as a cross-weight climb).
- The $k \leftrightarrow \nu$ dictionary (K872/K873) — an artifact of comparing a WEIGHT (which space) to a POSITION (which state) across domains. Not needed in one domain — I retire K873’s $k+\nu=g/2$ as a patch on a non-problem.
- The two-point across-domain Bergman distance (2.666 nats, needing the FK book) — there is no across-domain distance in one domain.

What it KEEPS (one-domain linear algebra)

- ONE Hilbert space: $H^2(D_{IV}^5)$, the Bergman space of the single domain (Born = Bergman).
- THREE generation STATES $|\psi_e\rangle, |\psi_\mu\rangle, |\psi_\tau\rangle$ = three VECTORS in that one space, supported at the three ρ -strata of the ONE domain: $\rho_1=5/2$ (continuum), $\rho_2=3/2$ (threshold), 0 (Shilov).
- ONE condensate operator $\hat{O}$ (or vector $|O\rangle$) on H^2 .
- THE object = ONE 3×3 mass MATRIX $M_{ij} = \langle \psi_i | \hat{O} | \psi_j \rangle$ — this IS F585, read as linear algebra on one space.
- Masses = EIGENVALUES of M ; mixing = off-diagonal / eigenbasis misalignment. Pure linear algebra.

Why this dissolves the K872 coordinate inconsistency

The k -vs- ν conflict was comparing a Bergman WEIGHT (which weighted space) to a POSITION (which state). In ONE fixed space there is ONE coordinate. The generations are 3 vectors; there is no across-domain distance to define, no dictionary to source. The problem reduces to: write M , diagonalize it. Eigenvalues, not a ladder. The 2-day non-convergence was (per Grace’s catch) mixed coordinates — and the one-domain frame removes the mixing at the root.

The muon value, restated in linear algebra (the sharp remaining question)

m_μ = an EIGENVALUE of the 3×3 M on $H^2(D_{IV}^5)$. Does M — built from the 3 ρ -strata states + the condensate operator, all in the ONE domain — have eigenvalue ratios 1 : 207 : 3477? No tower, no book-distance. - The rank of M (= rank of $\hat{O}$) is still THE question (K869): one condensate $\rightarrow$ rank 1 $\rightarrow$ one mass. So the honest crux is now cleanly stated: what is $\hat{O}$ as an OPERATOR on $H^2(D_{IV}^5)$ — its rank and spectrum? - This is Casey-standard linear algebra: one operator, one space, eigenvalues. No FK-book weighted-Bergman-tower, no $k \leftrightarrow \nu$ dictionary, no across-domain distance.

What is UNAFFECTED

- The structural result is already one-domain: the 3 Wallach strata are strata OF the single D_{IV}^5 (the Wallach set is a property of the one domain). “3 ρ -support-regions / Wallach phases of the ONE D_{IV}^5 ” — why-3, no-4th, Principle #16. The paper (F676) stands; just phrase it as one domain’s phases, not a tower.
- The scale (α -tower) and shared coupling: separate, nearly-derived (K871) — unaffected.

Directions (redirect off the tower)

- ★ LYRA/GRACE: STOP sourcing the $k \leftrightarrow \nu$ dictionary / the across-domain two-point distance (retired). The object is the ONE 3×3 mass matrix $M = \langle \psi_i | \hat{O} | \psi_j \rangle$ on $H^2(D_{IV}^5)$. Specify $\hat{O}$ as a linear operator on the one space (the condensate coupling), and its rank + spectrum. Diagonalize. Do the muon as an eigenvalue, in the one domain, with linear algebra.
- ★ The real question (K869, now sharp): what is $\hat{O}$ and its rank? Rank-1 (one condensate) $\rightarrow$ one mass + off-diagonal seesaw; higher rank $\rightarrow$ more diagonal masses. This is the condensate-rank question, now cleanly one-domain-linear-algebra.
- GRACE: the structural paper — phrase as the ONE domain’s Wallach phases (not a nested tower). ELIE: the eigenvalue/rank cross-check on M, one domain.
- KEEPER: K872/K873 (coordinate dictionary) retired as artifacts of the imported tower; the object is the one 3×3 M on $H^2(D_{IV}^5)$; the crux is $\hat{O}$ ’s rank + spectrum.

— Keeper K874, 2026-07-24. Casey steer “linear algebra, one D_{IV}^5 ” DISSOLVES the K872 coordinate mess: the $k \leftrightarrow \nu$ dictionary / Bergman-weight ladder / across-domain 2.666-nat distance / nested tower were IMPORTED multi-domain machinery. ONE domain: $H^2(D_{IV}^5)$, three generation VECTORS at the ρ -strata, ONE 3×3 mass matrix $M = \langle \psi_i | \hat{O} | \psi_j \rangle$ (=F585), masses = EIGENVALUES, diagonalize. Retires K873’s $k + \nu = g/2$ (patch on a non-problem). Crux now sharp: what is $\hat{O}$ on H^2 and its RANK+spectrum (K869)? Structure already one-domain (Wallach phases of D_{IV}^5), paper stands. See [[Keeper_K872_RATIFY_alpha_ladder_unification_Lanes1and3_are_one_machine_and_Grace_coordinate_inconsistency_catch_07-24]], [[Keeper_K869_the_Lyra_Grace_divergence_is_ONE_physical_question_condensate_rank_1_seesaw_vs_rank_3_diagonal_07-24]], [[Lyra_F585_CKM_texture_forcing_overlap_matrix_between_support_strata_sqrt_mass_ratio_derived_vs_structural]], Casey (linear algebra, one D_{IV}^5 — standing order).